

Tom Hu

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EDUCATION

Western University, Master of Engineering – Software Engineering

Sep. 2024 – Aug. 2025 | London, ON

University of Windsor, Bachelor of Computer Science (Honours)

Sep. 2020 – Jan. 2024 | Windsor, ON

SKILLS

- **Languages:** Python, Java, C/C++, JavaScript, TypeScript, Unity, Swift, HTML5/CSS3
- **Frameworks & Libraries:** React, Next.js, Node.js, Express.js, Spring Framework, Django, FastAPI, TensorFlow, PyTorch, TensorRT
- **Databases & Cloud:** MySQL, PostgreSQL, MongoDB, Redis, SQLite, AWS (EC2, RDS, S3, EKS, Lambda), GCP
- **Tools & OS:** Nginx, Kafka, RabbitMQ, Kubernetes, Docker, JMeter, CI/CD, Git, Postman, Linux

EXPERIENCE

Software Engineer, Baimeng Information Technology Co., Ltd.

Sep 2025 – Dec 2025 | Shanghai, China

- Developed a real estate analysis and decision support system with **Django** and **React**, visualized data using **Chart.js**, optimized access and caching with **PostgreSQL** and **Redis**, and implemented fine-grained permissions via **Django REST Framework**.
- Implemented Chart.js chart export using **jsPDF** and **html2canvas**, dynamically adjusted capture areas and scaling, applied paginated rendering, and ensured charts rendered clearly and fully in PDFs across varying resolutions and DPI.
- Optimized **Django ORM** queries by applying `select_related` and `prefetch_related` to resolve N+1 query issues, reduced page load time from 3s to 113ms, and eliminated key performance bottlenecks.
- Designed and maintained unit tests, API tests, and frontend component tests, covering 120+ **Django** APIs, 80+ **Celery** tasks, and 50+ **React** components to guarantee system stability and reliability under high concurrency.
- Conducted team code reviews and managed version control with **Git**, enforced project coding standards and design principles, and ensured naming consistency, clear documentation, structured code, and single-responsibility functions.

Software Development Intern, Trip.com

Feb 2024 – Aug 2024 | Shanghai, China

- Designed backend services with **FastAPI** and **Celery**, integrated **MySQL**, **Redis**, and **Kafka** to support hotel search, booking, and order processing, and implemented hotel search and display features on the frontend with **React**.
- Optimized hotel listing queries handling 50–80 entries during peak periods by separating `COUNT` and pagination queries, applied **MySQL** composite indexes, sharding and replication strategies, and stabilized API response under 300ms from an initial 2.5s.
- Optimized order event throughput with **Kafka** by partitioning Topics, enabling batch asynchronous production, and controlling consumer concurrency, ensuring stable processing of 20,000–50,000 daily order events during peak periods.
- Encapsulated 10+ reusable hotel search, listing, and pagination components using **Material-UI**, standardized state and interaction logic, and deployed them via **CI/CD** to the internal npm registry, improving frontend code reuse and delivery stability.
- Enhanced bundle performance with **webpack-bundle-analyzer**, loaded core libraries via **CDN**, applied component-level lazy loading and **Tree Shaking**, reducing first-screen load time from 6s to approx. 825ms.

Software Developer (CO-OP), ChengHua

Jul 2023 – Aug 2023 | Chengdu, China

- Built a high-availability distributed environmental monitoring management system using a microservices architecture with **Spring Cloud**, **RabbitMQ**, and **MongoDB**, and integrated **React**, GIS maps, and **Recharts** for data visualization.
- Designed and implemented a role-based access control (**RBAC**) system using **Spring Security** and **MySQL** to manage user roles and permissions, resolved permission conflicts and unauthorized access, and secured critical order data and operations.
- Developed an asynchronous messaging system with **RabbitMQ**, applied producer-consumer decoupling, dead-letter queues, and up to three retries to ensure reliable and ordered delivery of monitoring data and notifications in the online monitoring system.
- Led optimization of large file uploads, implementing chunked upload, resumable transfer, and instant upload for files over 40MB, supported by **Spring Boot** for storage and recovery, reducing upload time from 9–18s to 1–2s.
- Set up **CI/CD** pipelines with **Docker** and **Jenkins**, leveraged **Kubernetes** for load balancing and auto-scaling, shortened release cycles, improved deployment stability and efficiency, and ensured high availability of the microservices architecture.

PROJECTS

Cloud-Based Social Media App

- Containerized backend microservices on **GCP** using **Kubernetes (GKE)**, optimized **Cloud SQL** indexes and connection pools to handle 10,000+ active daily requests stably, resolving service instability during peak periods.
- Managed multi-environment configurations via **YAML**, optimized **GKE** Pod scheduling and rolling updates using **Ingress**, **Service**, and load balancing, reducing production deploy time from 20min to 5min while ensuring environment consistency.
- Integrated **Firestore Authentication** and decoupled services, optimized authentication with session caching and asynchronous verification, limited daily login failures to under 10, and reduced identity-related incident tickets by 30 per week.

Online Computer Store Web Application

- Optimized **MongoDB** queries in a **Node.js** + **Express.js** backend to address product catalog loading delays, applied indexes and aggregation pipelines, and reduced single-page load time from 3s to 610ms, improving list responsiveness.
- Enhanced checkout and payment flow performance by optimizing **RESTful API** response logic, leveraging asynchronous requests and caching, and processing payments asynchronously with **Square Web Payments SDK**, keeping payment confirmation time around 1s.
- Deployed the online computer store application on **AWS**, used **S3** and **CloudFront** to cache product images, and applied Auto Scaling and load balancing to backend services, ensuring stable page and API performance under high concurrency.